ACADEMIC VISUAL REFERENCE · DATA MINING FOR COMPETITIVE ADVANTAGE · FOUNDATIONS → PATTERNS → DECISIONS

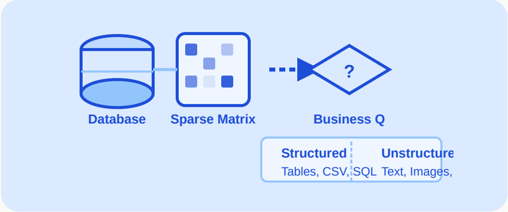
DATA MINING FOR COMPETITIVE ADVANTAGE — VISUAL CHEAT SHEET

6 Core Analysis Techniques · Business Datasets · Decision Criteria · Tips & Tricks | **Created by Johannes January** · Quinnipiac University

START HERE — Foundations & Business Understanding → EXPLORE — Exploratory Data Analysis → SIMPLIFY — Dimensionality Reduction → RECOMMEND — Collaborative Filtering → SEGMENT — Clustering Analysis → DISCOVER — Association & Sequence Mining

▶ START HERE · STEP 1 OF 6

Foundations & Business Understanding



BUSINESS CASE DATASET

Multi-Domain Business Data
Sales records, CRM exports, transaction logs — any structured or semi-structured source used in the modules below

BUSINESS QUESTIONS

- **WHY** Why does this business problem require data mining vs. simple reporting?
- **HOW** How do we translate a vague goal into a measurable data mining task?
- **WHAT** What KPIs (conversion, retention, lift) define a successful outcome?

KEY CONCEPTS

Structured Data · Unstructured · Sparsity · Scale · Noise

FRAMING PROCESS

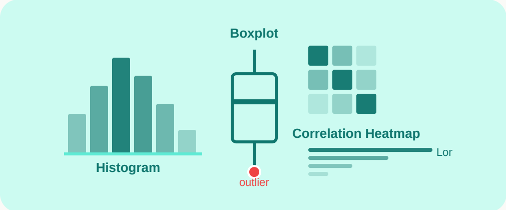
- Define the business question first
- Identify data sources & quality
- Map decision to measurable KPI

💡 **TIP**
"Always define the decision before the algorithm."

⚡ **TRICK**
Translate vague goals into measurable targets — conversion, retention, lift.

🔍 EXPLORE · STEP 2 OF 6

Exploratory Data Analysis (EDA)



BUSINESS CASE DATASET

NYC Property Sales Dataset
NYC Dept. of Finance — rolling sales data with borough, neighborhood, building class, sale price & date

BUSINESS QUESTIONS

- **WHY** Why are property prices distributed so unevenly across boroughs and neighborhoods?
- **HOW** How do building class, size, and sale timing affect price distributions?
- **WHAT** What features show the strongest correlations with sale price?

KEY VISUALIZATIONS

Histogram · Boxplot · Heatmap · Bar Chart

DIAGNOSTIC HINTS

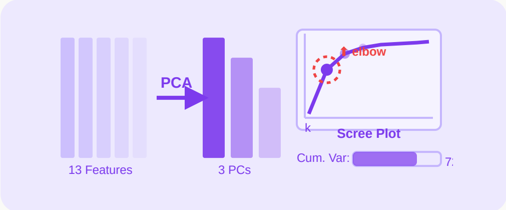
- Detect skewed distributions
- Dominant vs long-tail categories
- Check correlations before modeling

💡 **TIP**
EDA shows what customers actually do — not what you expect.

⚡ **TRICK**
Extreme outliers = VIP customers or fraud, not just "bad data."

📐 SIMPLIFY · STEP 3 OF 6

Dimensionality Reduction (PCA)



BUSINESS CASE DATASET

Wine Quality Dataset (UCI)
13 chemical features (acidity, sulphates, alcohol…) across red & white wines — ideal for reducing correlated chemical dimensions

BUSINESS QUESTIONS

- **WHY** Why do 13 chemical features create redundancy when predicting wine quality?
- **HOW** How many principal components capture 80%+ of the variance in wine profiles?
- **WHAT** What hidden quality dimensions emerge from the compressed chemical space?

TECHNIQUE

PCA · Scree Plot · Cumul. Variance · 2D Scatter

SELECTION CRITERIA

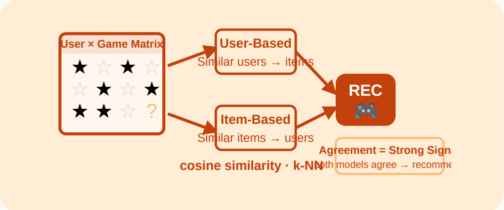
- Scree plot elbow point
- Variance threshold: 70–90%
- Interpretability vs compression

💡 **TIP**
PCA simplifies patterns but hides individual feature meaning.

⚡ **TRICK**
Use PCA for modeling efficiency — not for final business storytelling.

🎯 RECOMMEND · STEP 4 OF 6

Collaborative Filtering



BUSINESS CASE DATASET

Google Play Store Games Dataset
Kaggle — user ratings, installs, category & review data for mobile games; sparse user-game rating matrix

BUSINESS QUESTIONS

- **WHY** Why do users engage with certain games but not others with similar features?
- **HOW** How do we use neighborhood ratings to fill in a sparse user-game preference matrix?
- **WHAT** What games should be recommended to maximize installs and session time?

TECHNIQUES

User-Based CF · Item-Based CF · Cosine Sim

SELECTION CRITERIA

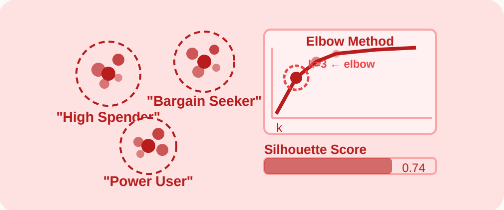
- k neighbors & similarity threshold
- Precision vs coverage balance
- Agreement between both engines

💡 **TIP**
Item-based CF scales better and is more stable over time.

⚡ **TRICK**
When user-based and item-based agree → that's your strongest recommendation.

🔵 SEGMENT · STEP 5 OF 6

Clustering Analysis



BUSINESS CASE DATASET

Credit Card Customer Transaction History
Kaggle — 8,950 customers with balance, purchase frequency, cash advance & payment behavior across 18 behavioral features

BUSINESS QUESTIONS

- **WHY** Why do credit card customers exhibit fundamentally different spending behaviors?
- **HOW** How do purchase frequency, balance, and cash advance patterns define natural groupings?
- **WHAT** What distinct customer personas can marketing act on for targeted campaigns?

TECHNIQUES

K-Means · Hierarchical · Dendrogram

SELECTION CRITERIA

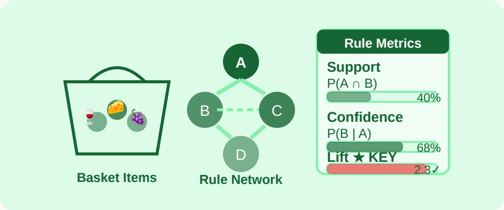
- Elbow method (within-cluster SSE)
- Silhouette score validation
- Stakeholder interpretability

💡 **TIP**
The "best" cluster count is the one stakeholders can act on.

⚡ **TRICK**
Rename clusters as personas — "Bargain Seekers", "Power Users" — to drive adoption.

🔗 DISCOVER · STEP 6 OF 6

Association & Sequence Rule Mining



BUSINESS CASE DATASET

Online Retail Store Transactions (UCI)
541,909 transactions from a UK-based online retailer — invoice, stock code, quantity, customer ID & country; ~4,000 unique products

BUSINESS QUESTIONS

- **WHY** Why do certain products get purchased together more often than by chance alone?
- **HOW** How frequently do product combinations co-occur across all transactions (support)?
- **WHAT** What cross-sell pairings will reliably increase average basket size and revenue?

TECHNIQUES

Apriori · Assoc. Rules · Seq. Mining

SELECTION CRITERIA

- Min. support (frequency filter)
- Confidence: P(B|A) reliability
- Lift > 1: true association strength

💡 **TIP**
High lift reveals influence — not just volume.

⚡ **TRICK**
Use high-lift rules to trigger targeted recs — not store-wide promotions.

⚠ COMMON CHALLENGES

- **Sparsity** — Most user-item cells are empty
- **Cold Start** — New users/items have no history
- **Popularity Bias** — Popular items dominate signals
- **Interpretability** — Performance vs explainability
- **Full Pipeline** — EDA → PCA → Cluster → CF → Rules

Exploration → Structure → Patterns → Prediction → Business Decisions
Data moves through each stage to generate competitive advantage.

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